

AI ENGINEER CAREER ROADMAP

Student → AI Engineer | Foundation: Web Development | Target Timeline: 3 Months

CURRENT POSITION

Profile: Student, early-career, cross-domain generalist (coding, research, writing, strategy).
Core skill: Web Development (HTML/CSS/JS + likely backend/API basics).
Pattern: Comfortable exploring broadly; strongest growth lever now is converting breadth into one area of depth.

TARGET GOAL

Goal: Become an AI Engineer in 3 months.
Realistic entry point: Applied AI / LLM Engineer - building AI-powered products, not AI research.
Why this works: Your web-dev base lets you ship full AI applications fast, instead of starting from zero in deep ML theory.

SKILL GAP ANALYSIS

Area	Have	Gap	Priority
Programming	JS / web fundamentals	Python fluency, OOP	High
AI/ML Core	None yet	ML basics, embeddings, LLM concepts	High
GenAI Tooling	API-consumption instinct	LLM APIs, RAG, LangChain/LlamaIndex	High
Deployment	Web hosting basics	FastAPI, Docker, vector DBs	Medium
Systems/Depth	Broad exploration habit	One focused vertical, repeatable workflow	Medium

LEARNING PLAN

Mo.	Focus
1	Python fluency + Git; AI/ML foundations (vectors, embeddings, tokens); LLM APIs (Claude/OpenAI); prompt engineering basics.
2	RAG pipelines; vector DBs (Chroma/Pinecone); LangChain or LlamaIndex; intro scikit-learn for core ML literacy.
3	AI agents; FastAPI + Docker deployment; polish portfolio; mock interviews; applications.

SUGGESTED PROJECTS

- RAG Chatbot - AI assistant answering questions over your own notes/docs; deploy as a live web app (web-dev + AI combo).
- AI Agent Tool - task-specific agent (e.g., resume analyzer or research summarizer) using an LLM API + simple tool calls.
- Portfolio Micro-SaaS - one polished full-stack AI product with a clean UI, README, and deployed demo link.

NETWORKING STRATEGY

- Post weekly "build-in-public" updates on LinkedIn (progress, learnings, demos).
- Polish GitHub: pin AI projects with clear READMEs and live demo links.
- Join AI/LLM developer communities (Discord, X/Twitter) and engage genuinely.
- Enter 1-2 AI hackathons for fast feedback loops and visibility.
- Reach out directly to AI engineers/founders for short informational chats.

MONTHLY MILESTONES

Mo.	Milestone
1	Python + AI fundamentals solid; first mini AI script shipped.
2	RAG project live; LinkedIn presence building; 2nd project started.
3	2-3 projects in portfolio; applying to roles; interview-ready.

IMMEDIATE NEXT ACTIONS (THIS WEEK)

- Set up Python + Git environment; finish a fast Python fundamentals crash course.
- Get an LLM API key (Claude/OpenAI); write a 20-line script calling the API.
- Post your first "Day 1 of becoming an AI Engineer" update on LinkedIn.
- Pick Project #1 (RAG chatbot) and sketch its scope this week.